

3ITRC3 Data Structures

Class Test 2

Duration – 70 Mins.

MM – 20

Note: All questions are compulsory and carry equal marks. Make suitable assumptions, if necessary.

- #1 Write an algorithm/procedure/code to convert an Infix expression into a postfix algorithm.
- #2 Give an algorithm for a singly linked list that reverses the direction of the links.
- #3 Let P be a pointer to the first node in a singly linked list and X an arbitrary node in the list. Write an algorithm to delete this node from the list. If $X = P$, then P should be reset to point to the new first node in the list.
- #4 You have a one-dimensional array. How could you maintain two stacks in it? Write the insertion/push procedure for the same.
- #5 What is a Deque? How it can be used to implement a Stack. Explain your answer.